$$A\vec{x} = \lambda\vec{x} \quad \text{——(1)}$$

$$A.A\vec{x} = A.\lambda\vec{x}$$

$$A^2\vec{x} = \lambda.(A\vec{x})$$

from (1)

$$A^2\vec{x} = \lambda(\lambda\vec{x})$$

$$A^2\vec{x} = \lambda^2(\vec{x})$$

$\therefore \lambda^n$ is eigenvalue of A^n:

$$\det(A - \lambda I) = \begin{bmatrix} 1-\lambda & \phi \\ \phi & -1-\lambda \end{bmatrix}$$

$$(1-\lambda)(1+\lambda) + 1 = 0$$

$$\lambda^2 = 2 \therefore \lambda = \pm\sqrt{2}$$

$$\lambda^{13} = \pm(\sqrt{2})^{13} = \pm 64\sqrt{2}$$